LexiForge — Keyterm Health for Gnani Prisma v3.0

Gnani gives you 200 keyterm slots per call; LexiForge measures what a candidate list does on your own calls — and proves whether it helped or hurt.

PROBLEM

- Prisma v3.0 accepts up to 200 `keyterms`, but the business glossary is far larger and changes. Someone must choose the subset — manually, and usually once.
- Vendor benchmarks show *relevant* keyterms help on public clips. They do not tell a customer whether a list helps on their noisy, code-mixed calls, or whether it degrades ordinary recognition.
- A wrong list is a silent failure: missed brand/product/scheme names break search, analytics and compliance; over-broad lists can displace known words.

WHAT WE BUILT

A Hinglish-banking **Keyterm Health service**: takes a glossary, call audio and reference transcripts; selects a candidate list; runs six arms on the same audio — baseline, selected, mismatched, hard-negative, oracle-200, static-head-200; scores term recall, insertions, WER and paired deltas; returns a report with an A/B verdict and provenance.

A = critical terms (brand, product, must-have). **B** = all-real guardrail: keyterms must not wreck ordinary recognition.

EVIDENCE — SYNTHETIC 4-CALL PILOT

The selected arm was brand-only, 8–9 terms — not a 200-term list. "Fictional-name recall" means invented glossary names, not real bank brands. Output format: `transliterate`.

Arm	Real recall	Fictional-name recall	Insertions / utt
baseline	21/23 (91.3%)	2/12 (16.7%)	0.00
selected (brand-only, 8–9)	16/23 (69.6%)	10/12 (83.3%)	0.05
mismatched-200	17/23 (73.9%)	3/12 (25.0%)	0.10
oracle-200	19/23 (82.6%)	**12/12 (100%)**	0.10

Full six-arm table: docs/pilot4-results.md. Internal harness validation — synthetic TTS, not customer evidence.

- **H1** (mismatched < baseline real recall): −17.5pp, p = 0.062 — not rejected.
- **H2** (mismatched > baseline insertions): +0.10/utt, p = 0.503 — not rejected.
- **Pilot verdict** (selected vs baseline, B guardrail): real recall −22.5pp, CI [−36.7, −8.3] — **FAIL** under the −5pp margin.
- **Positive control:** oracle − mismatched +9.17pp, CI [0.0, +18.3] — direction consistent, **CI touches zero**.
- At n=4 the smallest one-sided sign-flip p is 0.0625; descriptive only.

Key finding: every keyterm list lowered real recall versus baseline — including the oracle — while fictional-name recall moved as intended (16.7% → 83.3–100%). **Open question:** ASR bias or romanisation drift? The per-term diff decides. Gnani has not measured transliterated-output accuracy, and the matcher has no transliteration; Hindi terms rely on pre-written romanised aliases.

WHY GNANI

Gnani sells the 200 slots and the model. LexiForge measures what a list does on the customer's own audio: a POC accelerator, a glossary guardrail, and a measured capability for the keyterm feature Gnani already ships. Deployment alongside the customer's stack is a roadmap item; the pilot ran locally.

STATUS AND LIMITS

- Working end to end: Gnani REST + Batch clients, Timbre TTS, six arms, report, recording pack, 20 consistent scripts, 841 tests, 47-case matcher gold set.
- **Synthetic TTS only; no customer audio.** Internal harness validation.
- n=4 is underpowered; a 20-call run is required before any confirmatory claim.
- Batch queue instability observed; REST with chunking is the reliable transport. Calls over 30s are chunked, which can cut words and affect context — a confound.
- Pilot audio was TTS; telephony processing exists in the repo but was not verified as applied here. WER is format-sensitive and not quoted as recognition quality.

LINKS

Repo: github.com/v1r4t/LexiForge · Detail: docs/pilot4-results.md · Design: docs/design.md · Pre-registration: docs/preregistration.md · Real-audio intake: docs/real-audio-intake.md